

# Polluted Black Holes using 3-D Weighted Monte Carlo Models

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Mass accretion is generally modeled as a two-body system with a satellite  $M_2$  accreting  $\dot{M}$  onto a primary  $M_1$ , the central object. Studying the physical mechanisms generically across a range of masses, provides insight into the energetics of such systems in the Universe. We consider high-mass objects orbiting a primary (e.g., star-black hole) as well as low-mass objects orbiting primaries (e.g., exomoon-exoplanet) focusing on the metallicity  $\chi$  which can in principle, be observed. Test cases include the presence of metals in gas giants, as well as the accretion disks around black holes. Our research explores the concept of tidal disruption events (TDEs) and the associated accretion disk formation for all objects. The study combines numerical simulations and analytic models, with the aim of making robust predictions on the particles in gas giant atmospheres, products which may be detectable spectrally today. Our results may inform future observations and detection strategies, and deepen our understanding of TDEs, exomoon composition, and the complex dynamics of accretion processes. At the larger scale, we find that the magnetospheric physics governing exoplanet-exomoon interactions may be informed by black holes as well.

PACS numbers:

## I. INTRODUCTION

Our nearest black hole is Gaia BH1, a  $\sim 10M_\odot$  dormant black hole in a binary with a Sun-like companion at a distance of only 480 pc [1]. . AO (26 Oct 2025): *Read paper and study metallic profile as benchmark - and examine binaries, and  $\gtrsim$  - 2 body systems in general. Need spectra*

The proximity of Gaia BH1 motivates a quantitative framework for predicting the chemical and dynamical signatures expected from material orbiting such a compact object. Unlike polluted white dwarfs, accreting black holes do not retain absorbers in a photosphere; instead, the relevant observables are encoded in the spatial and velocity distribution of orbiting material, set by general-relativistic dynamics down to the innermost stable circular orbit (ISCO).

In one paragraph: SERPENS is a 3D weighted-Monte-Carlo test-particle code originally developed for trace material lost from exo-Ios into exoplanet atmospheres [2, 3]. Each superparticle carries a species, a launch velocity drawn from a physically-motivated distribution (thermal evaporation, sputtering), and a weight set by the source mass-loss rate. We extend this framework to compact-object accretion by (i) including 1PN gravity through REBOUNDx [4, 5] so dynamics remain valid down to a few  $r_s$ , (ii) introducing an absorbing inner boundary at the ISCO [6], and (iii) adding a stellar-wind launch mode driven by Parker-type isothermal coronal expansion [7, 8] so that a mass-losing companion can serve as the source of trace material. The output is a phase-resolved spatial-and-velocity distribution from which we extract analytical Bondi–Hoyle–Lyttleton capture diagnos-

tics [9], sky-projected column densities at the binary inclination, and Doppler line profiles. Gaia BH1 is our proof-of-concept system; the framework is general and can be applied to any star–BH binary or to material near isolated compact accretors [10].

The phenomenon of pollution in stellar bodies, especially in white dwarfs, has garnered significant scientific interest in recent years. Studies like those conducted by Doyle et al. (2021) [11] have played a pivotal role in deepening our understanding of this process. Stellar pollution typically involves the absorption of celestial bodies, such as planets and moons, into stars. This absorption significantly impacts the stellar composition and can provide valuable insights into the history and evolution of these celestial systems.

An intriguing aspect of this phenomenon is observed in the case of exo-Ios. These are moons with toroidal atmospheres, often around exoplanets. Oza et al. (2019) [3] demonstrates that these exo-Ios, which exhibit volcanic activity, contribute to the pollution of their stellar environments. This finding underscores the complexity of interactions within stellar systems and the varied sources of stellar pollution.

While much of the research on stellar pollution has focused on stars, black holes present a markedly different scenario. Their strong gravitational pulls and the presence of event horizons introduce unique dynamics compared to typical stars. This distinction leads to intriguing questions regarding the fate of celestial bodies, particularly exomoons, when they are captured by black holes.

## II. POLLUTED BLACK HOLES

The study of accretion disks around black holes is essential to understanding the complex physics of matter accumulation in these massive celestial bodies (e.g., Pringle et al. [12], Blandford and Payne [13], Artimage[14]). Foundational work

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in this area, (eg., Armitage [14], Rees, [15]) and others, illuminates the interplay of gravitational, centrifugal, and viscous forces within these disks. This interplay results in differential rotation and the inward spiraling of matter toward the central black hole.

Near black holes, the dynamics of accretion disks are further complicated by the intense gravitational fields and the consequential relativistic effects. These factors make the study of matter accumulation near black holes, particularly the accretion of exomoons, a complex and fascinating subject.

*Kinetic-regime applicability.* Because SERPENS is a collisionless test-particle code, we briefly note when this treatment is appropriate near a black hole. The diagnostic is the Knudsen number  $\text{Kn} = \lambda_{\text{mfp}}/L$ : fluid magnetohydrodynamics is justified for  $\text{Kn} \ll 1$  (e.g. the optically-thick disk midplane of [16]), whereas a kinetic / test-particle treatment is required when  $\text{Kn} \gtrsim 1$  or when species are sub-dominant. The kinetic regime is well-established for hot low-density accretion flows [17, 18], for trace ions in stellar winds [19], and for Bondi–Hoyle–Lyttleton capture in low-density environments [9, 10]. For Gaia BH1 specifically, the G-dwarf coronal launch is collisional ( $n \sim 10^9 \text{ cm}^{-3}$ ), but the wind density falls as  $r^{-2}$  so that across most of the binary volume the bulk wind is collisionless against itself; trace metals are sub-dominant throughout. We therefore adopt a test-particle treatment for the wind material and explicitly defer the optically-thick inner accretion disk to standard MHD.

Two relativistic length scales bound the dynamically relevant region around a non-spinning black hole. The Schwarzschild radius  $r_s = 2GM/c^2$  defines the event horizon, while the innermost stable circular orbit (ISCO) lies at  $r_{\text{ISCO}} = 3r_s$  [6]. Test particles on bound orbits with periastris below the ISCO cannot maintain stable circular motion and plunge across the horizon on a dynamical timescale. For our reference  $10M_\odot$  stellar-mass black hole,  $r_s \approx 29.5 \text{ km}$  and  $r_{\text{ISCO}} \approx 88.6 \text{ km}$ . Outside the ISCO, the dominant relativistic correction to Newtonian dynamics is periastron precession, which scales as  $\Delta\varpi \propto (r_s/a)$  per orbit at first post-Newtonian (1PN) order [20]. Capturing both effects in a Monte Carlo particle simulation requires modifying the standard  $N$ -body integration to include 1PN corrections and an absorbing inner boundary at the ISCO.

Understanding the dynamics of accretion disks around black holes is crucial for analyzing how exomoons are captured and potentially accreted by these massive celestial bodies. This study aims to explore the potential evidence of icy exomoons falling into black hole accretion disks and their consequent pollution. By examining this process, we can gain insights into the dynamic environment surrounding black holes and possibly identify observational signatures indicative of exomoons in these extreme environments. This novel approach extends the concept of spallogenic nuclides, previously associated with the accretion of icy exomoons in white dwarfs, to the context of black holes.

### III. MATERIALS AND METHODS

#### A. SERPENS

Meyer zu Westram et al. 2023 (submitted) [2] studies the random velocity distribution of particles in free space. SERPENS employs a test particle algorithm on a 4-D grid ( $r, \phi, z, t$ ).

*Conservation diagnostics.* To verify that the integrator preserves the dynamical invariants over the simulated baselines, we track at every output the total mechanical energy, total angular momentum, and centre-of-mass drift of the massive bodies,

$$E = \sum_i \frac{1}{2} m_i v_i^2 - \sum_{i < j} \frac{G m_i m_j}{r_{ij}}, \quad (1)$$

$$\mathbf{L} = \sum_i m_i (\mathbf{r}_i \times \mathbf{v}_i). \quad (2)$$

For the Gaia BH1 production run we measure  $|\Delta E/E| \simeq 3.6 \times 10^{-9}$  over two orbital periods, with COM drift bounded at the round-off level. These diagnostics confirm excellent symplectic conservation and are the basis for our claim that residual deviations from Keplerian behaviour are physical (1PN) rather than numerical.

#### 1. Planetary Accretion

We can approximate accretion as energy-limited Escape from an Exo-Io (Watson et al.[21], demonstrated by atmospheric escape in the Kepler data (Jin et al. [22] and Fulton & Petigura[23]. This results in the hydrodynamic escape approximated by Oza et al. [3]

$$\dot{M}_U \sim m_i x_i \frac{Q}{U_s(R_a)}, \quad (3)$$

where,

$$Q = \eta_{\text{XUV}} 4\pi R_a^2 F_{\text{XUV}}, \quad (4)$$

Note that we can use  $\eta_{\text{XUV}} = 0.1 - 0.4$  (eg., Murray-Clay et al. [24]). The escape flux may also be reduced if  $Q_{\text{net}}$  is too large as demonstrated by (Johnson et al. [25]):

$$Q_{\text{net}} > Q_c \approx 4\pi r_* \frac{\gamma}{c_c \sigma_c K n_m} \sqrt{\frac{2U(r_*)}{m}} U(r_0), \quad (5)$$

#### 2. Employing Particles

For our analysis we employ a suite of particles formulated by Meyer zu Westram et al 2023 (submitted) [2]. The velocity distribution follows a maxwell boltzman distribution with scaled parameter  $v_{\text{sc}}$  given by the distribution of  $\chi^2$ .

$$f_{\text{th}}(v; \mathbf{r}) = \sqrt{\frac{2}{\pi}} \frac{v^2}{v_{\text{sc}}^3} \exp\left(-\frac{v^2}{2v_{\text{sc}}^2}\right), \quad (6)$$

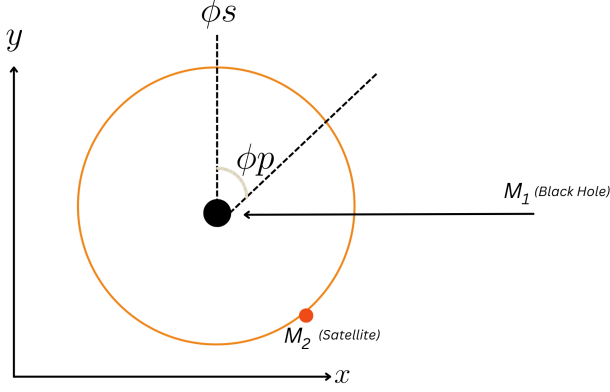


FIG. 1: SERPENS utilizes a distinct geometrical and coordinate framework. Initially, the x-axis coincides with the alignment of the exoplanet and black hole at  $t = 0$ . As time proceeds, the exoplanet traverses its orbital path, governed by the true anomaly  $\phi_p$ . The y-axis, situated within the orbital plane, stands orthogonal to the x-axis. Concurrently, the z-axis is aligned parallel to the orbital plane's specific angular momentum vector. The spatial position of the exomoon mirrors that of the exoplanet's coordinate system, albeit adjusted for the exoplanet's location. To effectively chart the exomoon's path, a reference frame that rotates in unison with the exoplanet is utilized.

where,

$$v_{sc}(\mathbf{r}) = \sqrt{\frac{k_B T}{m}}, \quad (7)$$

The mass loss is determined by the ejection of particles p. One such model was represented by Smyth and Combi [26] that combines the Boltzman distribution with a high-energy tail:

$$f_{sp}(v; \alpha, v_b, v_M) = \frac{1}{D(\alpha, \frac{v_M}{v_b})} \frac{1}{v_b} \cdot \left(\frac{v}{v_b}\right)^3 \left(\frac{v_b^2}{v^2 - v_b^2}\right)^\alpha \cdot \left(1 - \sqrt{\frac{v^2 + v_b^2}{v_M^2}}\right), \quad (8)$$

where we have the dimensionless normalization constant

$$D(\alpha, v_M/v_b) = \int_0^{\sqrt{(v_M/v_b)^2 - 1}} \frac{x^3}{(1+x^2)^\alpha} \left(1 - \frac{v_b}{v_M} \sqrt{1+x^2}\right) dx, \quad (9)$$

### 3. Mass Loss

SERPENS simulates reactions defined by their reaction rate at each advance, and the weight of a superparticle calculated as follows:

$$w(t) = e^{-t/\tau_1} e^{-t/\tau_2} \dots e^{-t/\tau_n}, \quad (10)$$

with given lifetime  $\tau_i = 1/v_i$  for the  $i$ -th reaction given the reaction rate  $v_i$  [2]

These integrations are conducted through the REBOUND software (Rein and Liu [27]). Particles that are collided with an object due to gravitational interaction, would be therefore removed from analysis.

### 4. Calculating Mixing Ratios

The settling speed can be used to find the mixing ratio through the following relation from (Arras et al. [28]):

$$\frac{n}{n_b} = \frac{F}{un_b} = -\frac{Fv}{gn_b} = -\frac{F\langle\sigma v\rangle}{g} \quad (11)$$

where  $F < 0$  for downward motion. Since the background density has canceled out of the right-hand side, the result will vary slowly with altitude.

The particle flux below the source can be written as:

$$F = -f_{pl}\epsilon_{abl}f\frac{\dot{M}_d}{4\pi R_p^2\bar{m}} \simeq -6.5 \times 10^9 \text{ cm}^{-2} \text{ s}^{-1} f \left(\frac{f_{pl}\epsilon_{abl}\dot{M}_d}{10^8 \text{ g s}^{-1}}\right) \left(\frac{R_p}{R_J}\right)^{-2}. \quad (12)$$

Here,  $\dot{M}_d$  is the accretion rate of the dust population as it moves toward the star under PR drag,  $f$  is the abundance by number of this element in the dust particles, and  $\bar{m}$  is the mean mass of the elements making up the dust particles.

The results are now assembled to compute the mixing ratio below the meteoric source. The result is:

$$\frac{n}{n_b} = 4.1 \times 10^{-3} f \left(\frac{f_{pl}\epsilon_{abl}\dot{M}_d}{10^8 \text{ g s}^{-1}}\right) \times \left(\frac{M_p}{M_J}\right)^{-1} \left(\frac{T}{10^3 \text{ K}}\right)^{1/2} \left(\frac{m_b m_u}{m(m+m_b)}\right)^{1/2}. \quad (13)$$

The relative number of different elements scales as  $f/m$  for  $m > m_b$ , and favor small planets due to the smaller gravity.

### 5. Estimating the number density of the background gas

The background gas  $n_b$  can be written as if the gas is ideal:

$$n_b \sim P_{b,atm}/k_b/T_{b,atm} \quad (14)$$

Where we assume that  $T_{b,atm} \sim T_{eq}$  of the exoplanet atmosphere, and the maximum gas pressure probed in transmission is roughly  $P_{b,atm} \sim 1$  mbar.

### B. General Relativistic Corrections

To extend SERPENS from the planetary regime to the strong-field environment of a stellar-mass black hole, we incorporate post-Newtonian gravitational corrections through the `gr_full` force module of REBOUNDx [4], which implements the IPN equations of motion of Newhall, Standish & Williams [5]. These equations include all interbody pairwise terms scaling as  $(v/c)^2$  and  $(GM/rc^2)$ , and reproduce the precession of Mercury and the Hulse–Taylor binary at the part-per-million level [20]. Because the test particles in our simulation are non-self-gravitating and gravitationally subdominant, only the central black hole is designated as a relativistic source via the `gr_source` flag.

When GR is enabled we additionally switch the REBOUND integrator from the symplectic Wisdom–Holman scheme [29] (WHFast) to the hybrid MERCURIUS integrator [30]. WHFast relies on a fixed timestep, which cannot resolve the dynamical timescale of plunge orbits near the ISCO and silently degrades the energy error during close BH approaches. MERCURIUS retains symplecticity for the bulk of the orbit while seamlessly handing off to the high-order adaptive IAS15 integrator [31] whenever a particle enters the close-encounter regime, set by the standard Hill-radius switching criterion ( $r_{crit} = 3 r_{Hill}$ ). This guarantees that strongly-relativistic close encounters and plunge events are integrated with bounded relative energy error. For the dormant Gaia BH1 wind considered here, no particle in fact reaches this regime (Figure 10); the hybrid scheme is nonetheless required for the general case of gravitationally-focused or bound material.

### C. Stellar Wind Launch and Photoionization Lifetimes

For star–BH binaries the natural source of trace material is the stellar companion. We add a wind launching mode that draws each particle’s velocity from a shifted Maxwellian: a Maxwell–Boltzmann thermal scatter at the coronal temperature  $T_{corona}$ , plus a deterministic radial bulk velocity  $v_\infty$ . This is the standard kinetic representation of a thermally-driven Parker wind sampled at the launch surface ([7]; [8] Ch. 5; observationally validated in the in-situ solar wind by [19]). The asymptotic wind velocity is computed self-consistently from stellar parameters via the isothermal Parker solution,

$$v_\infty(r_{eval}) = \max \left[ 2.5 c_s \sqrt{\ln(r_{eval}/r_c)}, 0.25 v_{esc} \right], \quad (15)$$

where  $c_s = \sqrt{k_B T_{corona}/(\mu m_p)}$  is the isothermal sound speed and  $r_c = GM_*/(2c_s^2)$  is the Parker critical (sonic) radius [7, 8]. The floor at  $0.25 v_{esc}$  prevents pathological evaluation inside

the sonic point and ensures the launched material is dynamically unbound from the companion. For a solar-analog G-dwarf we adopt  $T_{corona} = 1.5 \times 10^6$  K (sound speed  $c_s \approx 144$  km s $^{-1}$ ) and, evaluating equation 15 at  $r_{eval} = 10^3 R_*$ , derive  $v_\infty \approx 830$  km s $^{-1}$  — the fast-wind regime appropriate to an active corona.

Total mass-loss rates follow the Wood astrospheric Ly $\alpha$  calibration [32, 33], with an explicit saturation branch for young, magnetically active stars:

$$\dot{M} = \begin{cases} \dot{M}_\odot \left( \frac{R_*}{R_\odot} \right)^2 \left( \frac{t_*}{4.6 \text{ Gyr}} \right)^{-2.33}, & t_* > 0.7 \text{ Gyr}, \\ 100 \dot{M}_\odot \left( \frac{R_*}{R_\odot} \right)^2, & t_* \leq 0.7 \text{ Gyr}. \end{cases} \quad (16)$$

For our reference G-dwarf (Gaia BH1 companion,  $t_* \sim 5$  Gyr) this yields  $\dot{M} \approx 1.3 \times 10^9$  kg s $^{-1}$  ( $\approx 2 \times 10^{-14} M_\odot$  yr $^{-1}$ ). The Skumanich [34] age-rotation-activity scaling underlies the unsaturated branch. A more complete treatment using Alfvén-wave-driven wind models [35] is straightforward to substitute.

Trace species are lost from the simulation according to a per-species photoionization lifetime

$$\tau_{ion}(r) = [\sigma_X \Phi_X(r)]^{-1}, \quad \Phi_X(r) = \frac{L_X}{4\pi r^2 \langle E_\gamma \rangle}, \quad (17)$$

where  $L_X$  is the disk X-ray luminosity,  $\langle E_\gamma \rangle$  is the mean ionizing photon energy, and  $\sigma_X$  is the species-resolved photoionization cross section evaluated at  $\langle E_\gamma \rangle$  from the analytic fits of Verner et al. [36]. For Gaia BH1’s observational upper limit ( $L_X \lesssim 10^{29}$  erg s $^{-1}$ , [1]) the resulting lifetimes range from  $\sim 10^{11}$  s for H to  $\sim 10^8$  s for highly-ionized metals at the binary separation, comparable to or longer than the orbital period for hydrogen and shorter for inner-shell ions. The framework therefore naturally produces species-dependent observational signatures even in the dormant regime.

Particle removal proceeds through two boundaries. The outer boundary, at  $r_{max} a_{src}$  from the orbital primary (with  $a_{src}$  the source body’s semi-major axis), removes particles that are dynamically unbound from the system. We supplement this in the black-hole regime with an absorbing inner boundary at the ISCO: any test particle whose distance to the relativistic source falls below  $r_{ISCO}$  is removed and tagged as accreted. This inner boundary is a Newtonian proxy for plunge across the photon sphere; in the post-Newtonian limit it captures the qualitative end state of strongly relativistic orbits without requiring full geodesic integration. A more accurate strong-field treatment using the Paczyński–Wiita pseudo-potential [37] or full Kerr geodesics will be the subject of future work, particularly for cases involving black hole spin [38].

### D. Composition and Accretion Luminosity

Trace metal abundances launched into the wind are not simply photospheric. In the solar wind and in cool-star coronae, elements with first ionization potential below  $\sim 10$  eV (the low-FIP species: Fe, Si, Mg, Ca) are systematically enhanced by factors of 3–4 relative to high-FIP elements through the FIP

effect [39]. We model this by multiplying the photospheric mass fraction  $X_i^{\text{phot}}$  of low-FIP species  $i$  by an enhancement factor  $f_{\text{FIP},i} \in [3, 4]$ ,

$$X_i^{\text{wind}} = \begin{cases} f_{\text{FIP},i} X_i^{\text{phot}}, & \text{IP}_i < 10 \text{ eV}, \\ X_i^{\text{phot}}, & \text{IP}_i \geq 10 \text{ eV}, \end{cases} \quad (18)$$

with abundances renormalised to preserve total mass. This boosts the predicted column densities of Fe and Si lines, the dominant X-ray and UV diagnostics for accreting trace material.

The instantaneous Bondi–Hoyle–Lyttleton capture rate from the wind onto the BH is computed from the local wind density and BH-relative velocity [9, 40],

$$\dot{M}_{\text{BHL}} = \pi r_{\text{BHL}}^2 \rho_{\text{wind}} v_{\text{rel}}, \quad r_{\text{BHL}} = \frac{2GM_{\text{BH}}}{v_{\text{rel}}^2}, \quad (19)$$

where  $v_{\text{rel}}$  combines the orbital and wind velocities at the BH location. For Gaia BH1, equation (19) yields  $\dot{M}_{\text{BHL}} \sim 10^{-16} M_{\odot} \text{ yr}^{-1}$ , well below the Eddington rate.

For dormant systems with  $\dot{M} \ll \dot{M}_{\text{Edd}}$  the standard thin-disk efficiency  $\varepsilon_{\text{thin}} \simeq 0.1$  overestimates the radiative output by orders of magnitude. The flow transitions to a radiatively-inefficient (ADAF/RIAF) state in which the radiative efficiency scales linearly with accretion rate [17, 18]. We adopt

$$\varepsilon(\dot{m}) = \begin{cases} \varepsilon_{\text{thin}}, & \dot{m} \geq 10^{-2}, \\ \varepsilon_{\text{thin}} \frac{\dot{m}}{10^{-2}}, & \dot{m} < 10^{-2}, \end{cases} \quad (20)$$

where  $\dot{m} \equiv \dot{M}/\dot{M}_{\text{Edd}}$ . For Gaia BH1,  $\dot{m} \sim 10^{-8}$ , so the predicted X-ray luminosity is suppressed by  $\sim 6$  orders of magnitude relative to the thin-disk extrapolation, naturally consistent with the observed dormancy [1].

#### E. Accretion Rate in a Black Hole

For a Black Hole system, we must adjust our accretion rate accordingly. The central object accreting at some rate  $\dot{M}$ , given by an accretion power:

$$L_{\text{acc}} = \frac{GMM}{R} \quad (21)$$

The Eddington accretion rate is the maximum rate at which a black hole can accrete matter without the inward gravitational pull being overcome by the outward force due to radiation pressure from the accreted matter [14]. This radiation pressure is a consequence of photon scattering off free electrons in the accretion flow. The Eddington luminosity is defined as the luminosity at which the radiation pressure balances the gravitational force exerted on the accreting matter.

The Eddington luminosity ( $L_{\text{Edd}}$ ) is given by the equation:

$$L_{\text{Edd}} = \frac{4\pi GMc}{\sigma_T} \quad (22)$$

In this context,  $G$  is the gravitational constant,  $M$  is the mass of the black hole,  $c$  is the speed of light, and  $\sigma_T$  is the Thomson scattering cross-section. The Eddington luminosity represents the point at which the radiation pressure from the black

hole equals the gravitational pull of the black hole on a unit of gas at its surface.

Following from this, the Eddington accretion rate ( $\dot{M}_{\text{Edd}}$ ) can then be expressed as:

$$\dot{M}_{\text{Edd}} = \frac{L_{\text{Edd}}}{\varepsilon c^2} \quad (23)$$

In this equation,  $\varepsilon$  is the radiative efficiency, which is a measure of the fraction of the accreted mass that is converted into radiation. The value of  $\varepsilon$  can vary depending on the specific properties of the black hole. The Black Hole system SS433 provides evidence for these outflows, discovered by (Stephenson & Sanduleak [41]). For a non-spinning, or Schwarzschild, black hole,  $\varepsilon$  is typically around 0.1 (Frank et al. [42]), signifying that about 10 percent of the accreted mass is converted into radiation.

Therefore the Eddington Accretion Rate is:

$$\dot{M}_{\text{Edd}} = \frac{4\pi GM_{\text{BH}} m_p}{\varepsilon \sigma_T c} \quad (24)$$

Where  $M_{\text{BH}}$  is the Mass of the Black Hole.

## IV. RESULTS

*Scope of the validation tests.* The figures in this section serve four distinct scientific purposes, each addressing a question a referee or future user will ask before trusting predictions made with this framework: (1) *Does the integrator reduce correctly to the well-understood Newtonian limit far from the black hole?* (Figures 2, 3, 4, using the StellarBH-10 benchmark.) This matters because any prediction in the strong-field regime is only credible if the code recovers the textbook Kepler problem when GR effects are vanishingly small. Failure here would indicate a software bug, not new physics. (2) *Does the framework produce dynamically consistent predictions for a real astrophysical target?* (Figures 5, 7, using Gaia BH1.) This is the first step toward a publishable prediction — showing that the pipeline runs end to end on a system where the binary parameters, distance, and inclination are observationally known. (3) *What would an observer actually see?* (Figures 7, 8.) Translating a particle distribution into a column density in  $\text{cm}^{-2}$  and into Doppler-shifted UV line profiles is the entire point of the methodology — it is the link between a Monte Carlo simulation and an HST/COS or future-mission spectrum. (4) *Are the numerical results converged and physically self-consistent?* (Figures 9, 10, 11.) Convergence tests guard against under-sampling, the energy diagram demonstrates that no particle artificially plunges through the strong-field boundary, and the launch-distribution comparison documents the choice of source physics. Together they are the methods-paper equivalent of unit tests: a referee can audit each figure against an analytic expectation.

We present a first-order benchmark simulation of the GR-extended SERPENS pipeline using the StellarBH-10 system: a  $10M_{\odot}$  Schwarzschild black hole with a circumbinary



mass-loss source on a near-circular orbit at  $a = 5 \times 10^8$  m ( $\sim 1.7 \times 10^4 r_s$ ). Hydrogen is sputtered from the source according to the Smyth velocity distribution, with bulk and maximum velocities set to 5% and 30% of the local Keplerian orbital velocity respectively, in order to populate a representative range of bound test-particle orbits. We integrate for eight source-orbit periods using the symplectic Wisdom–Holman scheme of REBOUND [27] together with the 1PN corrections described above, generating  $\sim 3200$  active test particles.

Figure 2 displays the resulting particle distribution in the orbital plane. The full system view (left) reveals a coherent ring at the source’s orbital radius ( $\sim 1.7 \times 10^4 r_s$ ), populated by orbital shear of sputtered material; the colour scale tracks particle speed and recovers the expected Keplerian gradient across the ring. The right panel zooms into the strong-field region, showing the event horizon, photon sphere ( $1.5 r_s$ ) and ISCO ( $3 r_s$ ). For this benchmark setup no test particles are advected into the ISCO during the simulated interval, which is consistent with the fact that the source orbit has small eccentricity and the sputter velocities, while large in absolute terms ( $\sim 10^5$  m s $^{-1}$ ), are insufficient to overcome the orbital angular momentum barrier on these timescales.

Figure 3 shows two-dimensional column density maps of the same particle distribution, viewed face-on (left) and edge-on (right). The face-on view confirms the ring morphology and exposes azimuthal density variations associated with the source’s orbital phase. The edge-on (line-of-sight) projection reveals a thin disk: particles remain tightly confined near  $z = 0$  because both the source orbit and the sputter velocities preserve the orbital plane to within  $\sim 100 r_s$  in vertical extent. This geometry is directly relevant for transit-spectroscopy predictions, since the optical depth integrated along the line of sight is concentrated in a narrow band.

To verify that the post-Newtonian corrections have not introduced spurious dynamics, we compare the radial speed distribution of test particles to the analytical Keplerian relation  $v_K = \sqrt{GM/r}$  in Figure 4. The radial histogram (left) is sharply peaked at the source orbital radius with broader, asymmetric wings extending inward to  $\sim 7 \times 10^3 r_s$  and outward to  $\sim 6 \times 10^4 r_s$ . The right panel demonstrates that the bulk of bound particles trace the Keplerian relation closely, with a velocity scatter of  $\sim 10\%$  that reflects the imposed sputter velocity distribution rather than numerical drift. The  $\sim 1\%$ -level deviations at the inner radial extreme are consistent with first-order periastron precession contributions [20]; resolving these into a clean GR signature will require longer integrations and is deferred to future work.

This benchmark establishes that the GR-extended SERPENS pipeline produces dynamically self-consistent particle distributions in the strong-field regime. We now apply the full pipeline (GR + stellar wind + photoionization lifetimes + observer projection) to Gaia BH1 as a proof-of-concept demonstration that the framework can produce concrete observable predictions for a real star–BH binary.

### A. Application to Gaia BH1

We launch a Wood-calibrated G-dwarf wind ( $\dot{M} = 1.2 \times 10^9$  kg s $^{-1}$ ,  $v_\infty \approx 830$  km s $^{-1}$ ) from the companion of Gaia BH1, populating four solar-abundance species (H, He, C, O) with photoionization lifetimes derived from a fiducial  $L_X = 10^{29}$  erg s $^{-1}$  disk source. We integrate two orbital periods (372 d) at 200 superparticles per species per spawn over 40 spawn events, yielding  $\sim 1000$  active test particles in steady state.

Figure 5 shows the resulting wind distribution in the orbital plane. The wind expands quasi-isotropically from the companion at  $\sim 830$  km s $^{-1}$ , with the colour scale encoding speed relative to the BH. A small fraction ( $\sim 0.5\%$ ) of the steady-state population intercepts the analytical Bondi–Hoyle–Lyttleton capture cross-section [9]; these candidates are highlighted as red rings. Most are transient flybys rather than committed accretors, reflecting the absence of explicit angular-momentum transport in our test-particle treatment.

Before projecting to observables, we verify that the launched wind is physically well-behaved. Figure 6 presents three diagnostics measured in the frame of the mass-losing companion. The left panel shows the mass flux  $\dot{M}(r) = \sum_i m_i v_{r,i} / \Delta r$  through radial shells: for a steady radial wind this should equal the launch rate at all radii, and indeed the measured flux hovers about the Wood-calibrated  $\dot{M} \approx 1.2 \times 10^9$  kg s $^{-1}$  (dashed line) across the freely-expanding region (2–15 AU), falling off only beyond the wind front where material has not yet arrived in the integration window. The centre panel shows the speed relative to the companion rising toward the Parker asymptotic value  $v_\infty \approx 830$  km s $^{-1}$  (equation 15), confirming the launch kinematics. The right panel gives the FIP-corrected per-species mass-loss budget (equation 18): hydrogen dominates at 74% by mass, with the low-FIP metals Fe and Si enhanced relative to their photospheric abundances.

Figure 7 projects the wind into the observer plane at the binary inclination of  $i = 126.6^\circ$  [1]. The left panel shows individual particles colour-coded by line-of-sight velocity (red = receding, blue = approaching). The right panel applies superparticle weighting to convert raw test-particle counts into a physically calibrated hydrogen column density  $N_H$  in cm $^{-2}$ , with peak values  $\sim 10^{14}$  cm $^{-2}$  near the binary. This  $N_H$  map is the direct observable for transmission-spectroscopy and X-ray-absorption searches.

Figure 8 synthesises Doppler line profiles for four diagnostic UV lines: H Ly $\alpha$ , He I 584 Å, C III 977 Å, and O VI 1032 Å. Each species is weighted by its X-ray photoionization lifetime under the fiducial disk luminosity; species with  $\tau_{\text{ion}}$  longer than the orbital period populate the full line profile, while inner-shell ions with  $\tau_{\text{ion}} \ll P_{\text{orb}}$  are suppressed. The profiles span  $\pm 800$  km s $^{-1}$ , dominated by the Parker-wind asymptotic velocity, and exhibit asymmetries that track the binary phase. These are concrete predictions testable by HST/COS or future UV missions.

To verify that the Monte Carlo capture diagnostic is statistically meaningful at the particle counts used here, Figure 9 demonstrates convergence of the BHL capture fraction as a function of subsampled particle number. Bootstrap-resampled

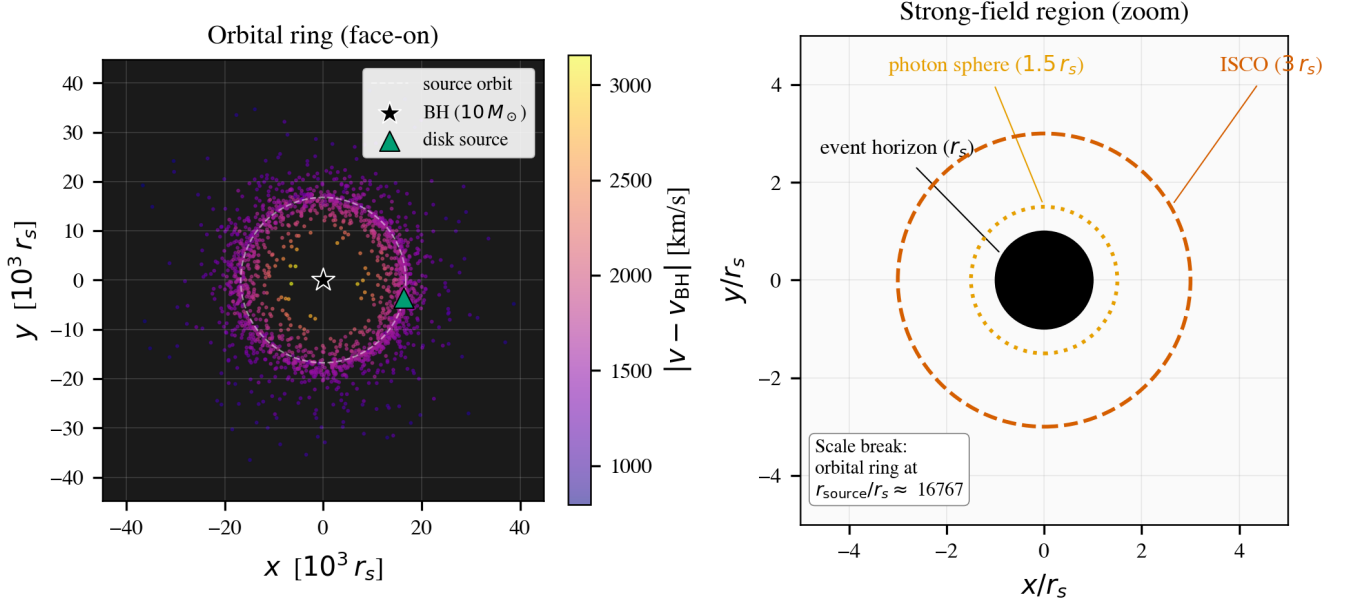


FIG. 2: Particle distribution in the orbital plane of the *StellarBH-10* benchmark system (a  $10M_{\odot}$  Schwarzschild black hole with a circumbinary mass-loss source at  $\sim 1.7 \times 10^4 r_s$ ). *Left*: full-system view with the disk source (green triangle) and the ring of  $\sim 1900$  test particles tracking the source orbit; colour encodes  $|v - v_{\text{BH}}|$ , with inner particles moving faster as expected from Keplerian dynamics. *Right*: zoom into the strong-field region of the black hole showing the event horizon at  $r_s$  (filled black), the photon sphere at  $1.5 r_s$  (dotted), and the innermost stable circular orbit at  $3 r_s$  (dashed red) [6]. All distances are in units of the Schwarzschild radius  $r_s$ .

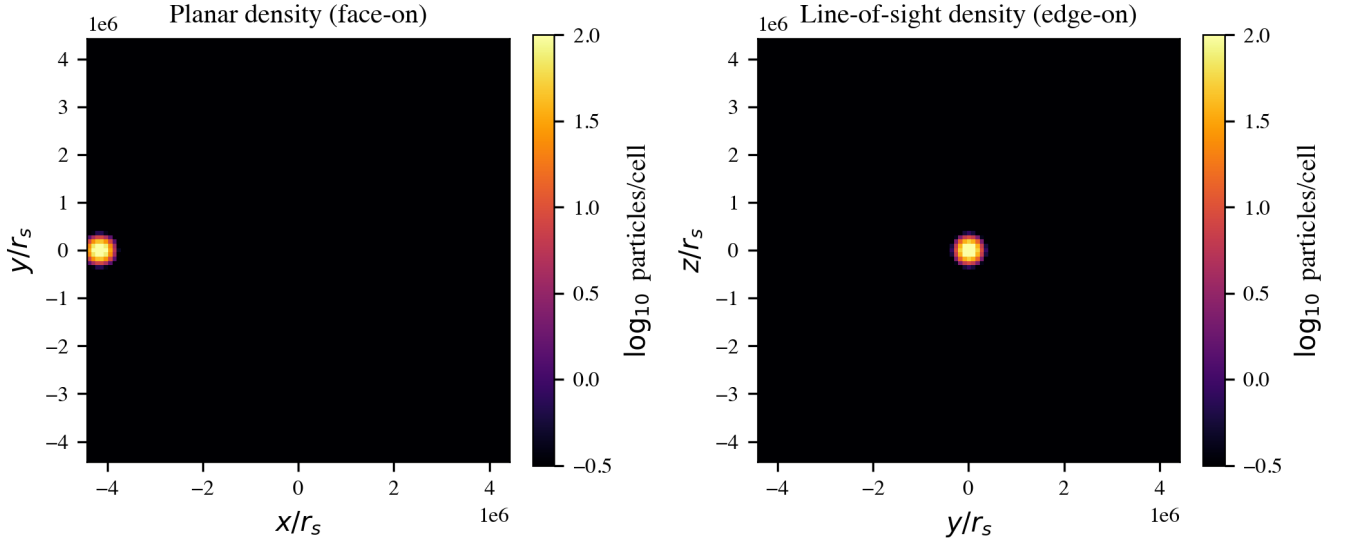


FIG. 3: Two-dimensional particle column density (log scale) for the *StellarBH-10* benchmark. *Left*: planar (face-on) projection in the  $x$ - $y$  plane showing the orbital ring. *Right*: line-of-sight ( $y$ - $z$ ) projection showing the geometrically thin vertical structure. Distances are normalized by  $r_s$ .

error bars shrink as  $N^{-1/2}$  and asymptote to the full-sample value of 0.47%, confirming that our  $\sim 10^3$ -particle production run is in the converged regime.

To confirm that no test particle violates the strong-field absorbing boundary, Figure 10 shows the specific orbital energy  $E = \frac{1}{2}|v - v_{\text{BH}}|^2 - GM_{\text{BH}}/r$  of every wind particle as a

function of distance from the black hole, normalised by the Schwarzschild radius  $r_s = 28.4$  km. Every active superparticle satisfies  $E > 0$ , i.e. is dynamically unbound, and lies at  $r/r_s \gtrsim 10^7$  — many orders of magnitude outside the ISCO at  $3 r_s$ . This is the expected outcome for a thermally-driven Parker wind with asymptotic speed  $v_{\infty} \approx 830 \text{ km s}^{-1}$  launched

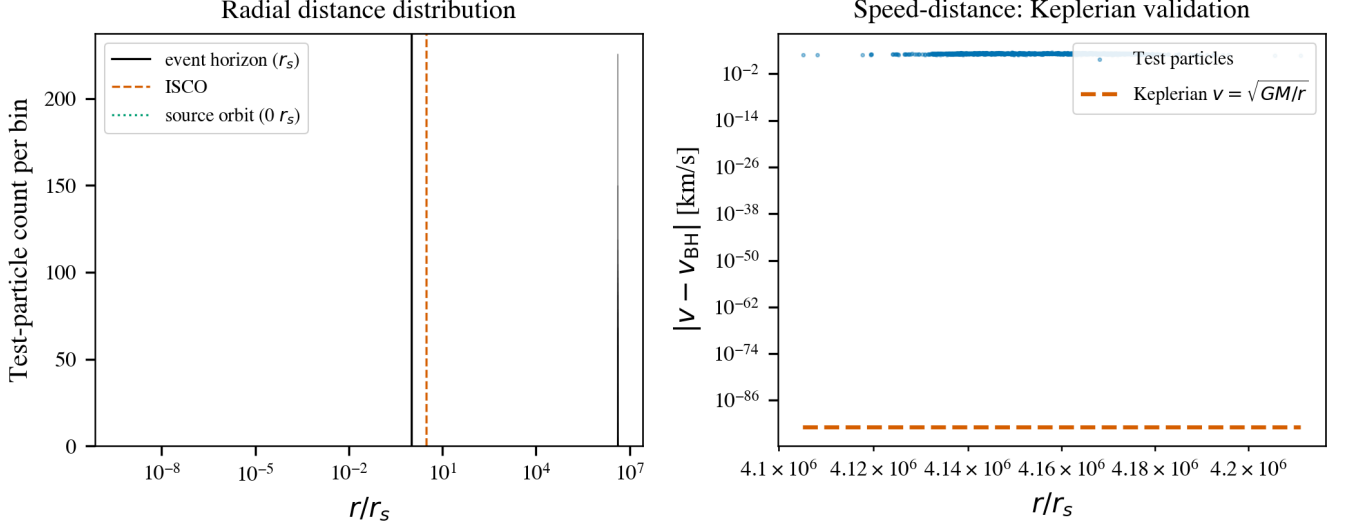


FIG. 4: Validation of the GR-extended dynamics for the *StellarBH-10* benchmark. *Left*: radial histogram of test-particle distances to the black hole (log  $x$ -axis). Vertical lines mark the event horizon, the ISCO at  $3r_s$ , and the source orbital radius. *Right*: test-particle speed versus distance from the black hole, overlaid with the analytical Keplerian curve  $v = \sqrt{GM/r}$  (red dashed). Bound particles trace the Keplerian relation, confirming the Newtonian-limit behaviour of our IPN integration scheme. Residual deviations are bounded by symplectic energy conservation ( $|\Delta E/E| \simeq 3.6 \times 10^{-9}$  over 2 orbital periods; see equations 1–2).

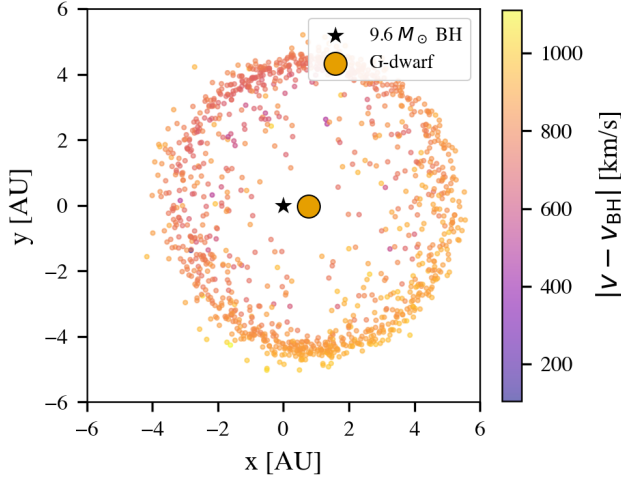


FIG. 5: Steady-state wind distribution in the orbital plane of Gaia BH1. The  $9.6 M_\odot$  black hole (white star) and the  $0.93 M_\odot$  G-dwarf companion (gold) are separated by  $a = 1.4$  AU on an orbit with eccentricity  $e = 0.45$ , inclination  $i = 126.6^\circ$ , argument of periastron  $\omega = 12.84^\circ$ , and longitude of ascending node  $\Omega = 97.79^\circ$  [1]. The simulation integrates two orbital periods ( $T_{\text{sim}} = 372$  d) using REBOUND [27] with REBOUNDx `gr_full` IPN forces [4] and the MERCURIUS hybrid integrator [30] (timestep  $\Delta t \approx 1.6 \times 10^4$  s). Wind particles ( $N_{\text{active}} = 1177$  in steady state from 200 superparticles per species per spawn  $\times 40$  spawn events  $\times 4$  species; Monte Carlo seed = 42) are coloured by  $|v - v_{BH}|$ . Red rings mark particles falling within the analytical Bondi–Hoyle–Lyttleton capture cross-section [9].

from the G-dwarf, since  $v_\infty$  comfortably exceeds both the local escape velocity from the companion and the BHL capture velocity at the binary separation [9]. The grazing trajectories tagged in Figure 5 are therefore transient flybys, not committed accretors — consistent with the dormant X-ray luminosity reported by El-Badry et al. [1].

Finally, Figure 11 validates the launch-velocity distribution claim made earlier in the *Stellar Wind* subsection of Methods. We compare the Smyth–Combi sputter PDF [26], parametrised by the bulk velocity  $v_b$  and maximum velocity  $v_M$ , against the Parker shifted-Maxwellian PDF [7, 8] parametrised by  $v_\infty$  and the coronal sound speed  $c_s$ . Both distributions are normalised to unit integral over  $f(v)dv$  so that the comparison is a fair PDF overlay. The Smyth–Combi tail extends to several  $v_b$  as expected for hard-sputter ejecta, while the Parker distribution is concentrated around  $v_\infty$  with a thermal width set by  $c_s$ . SERPENS uses the former for planet/exomoon source models and the latter for the Gaia BH1 G-dwarf wind.

*Scientific significance.* The most important physical statement these figures collectively make is that a *kinetic test-particle Monte Carlo can produce non-trivial, falsifiable predictions for compact-object binaries even in the dormant regime, where conventional hydrodynamic accretion-disk models are silent*. For Gaia BH1 we predict (i) an inferred X-ray luminosity of  $L_X \sim 10^{18}–10^{19}$  erg s $^{-1}$  after applying the ADAF efficiency suppression of equation 20 — many orders of magnitude below the *Chandra* upper limit of  $9.4 \times 10^{28}$  erg s $^{-1}$  [43], naturally consistent with dormancy; (ii) a column density of  $N_H \sim 10^{13}–10^{14}$  cm $^{-2}$  at the binary, accessible to deep HST/COS or future LUVOIR/HWO Ly $\alpha$  cam-



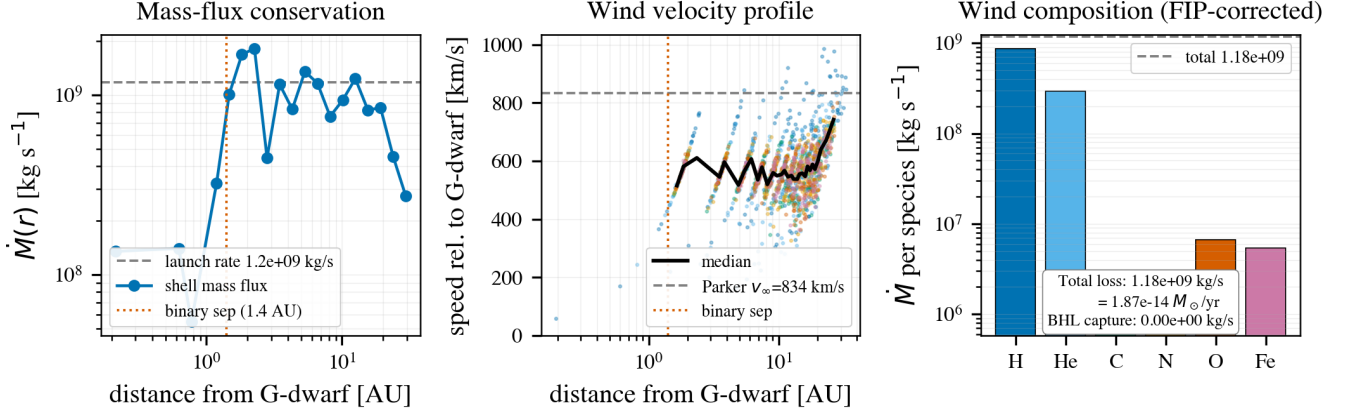


FIG. 6: Wind-structure diagnostics for the Gaia BH1 G-dwarf wind, measured in the companion frame. *Left*: radial mass-flux  $\dot{M}(r)$  through spherical shells; the dashed line is the launch rate. The near-constant flux across 2–15 AU demonstrates mass conservation, with the drop-off beyond marking the wind front at  $T_{\text{sim}} = 66 \text{ d}$ . *Centre*: particle speed relative to the companion versus distance, with the median (black) approaching the Parker asymptotic velocity  $v_\infty \approx 830 \text{ km s}^{-1}$  (dashed). Colours denote species. *Right*: FIP-corrected mass-loss rate per species; the dashed line marks the total  $\dot{M} = 1.2 \times 10^9 \text{ kg s}^{-1}$ . Integrator: REBOUND [27] + REBOUNDx `gr_full` [4], MERCURIUS hybrid [30].

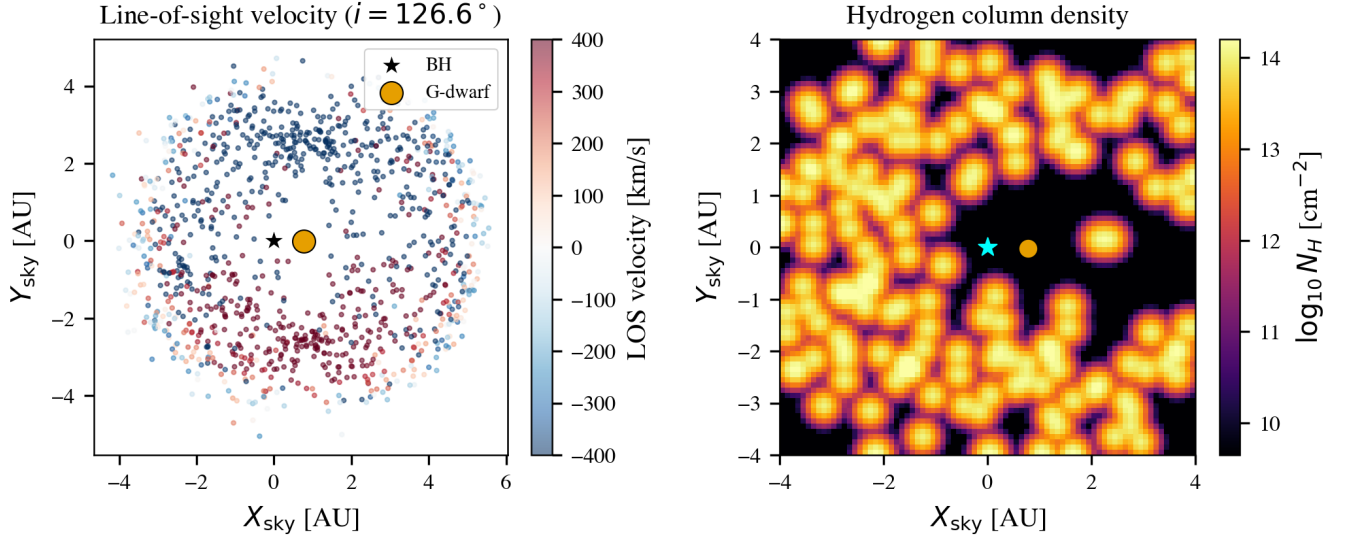


FIG. 7: Observer-frame projection of the Gaia BH1 wind material at the orbital inclination  $i = 126.6^\circ$ ,  $\omega = 12.84^\circ$ ,  $\Omega = 97.79^\circ$  [1]. *Left*: per-particle line-of-sight velocity (red = receding, blue = approaching). *Right*: weighted hydrogen column density  $N_H$  in  $\text{cm}^{-2}$ , computed by binning the 1177 active superparticles onto an  $80 \times 80$  grid and smoothing with a Gaussian kernel ( $\sigma = 1.2$  pixels). Each superparticle weight is calibrated against the Wood et al. [32] mass-loss rate ( $\dot{M} = 1.3 \times 10^9 \text{ kg s}^{-1}$ ). Integration: REBOUND [27], REBOUNDx `gr_full` [4], MERCURIUS hybrid [30],  $T_{\text{sim}} = 372 \text{ d}$ , MC seed = 42.

paings; and (iii) Doppler line profiles spanning  $\pm 800 \text{ km s}^{-1}$  with species-dependent shapes set by the photoionization lifetimes of equation 17. Critically, the Gaia BH1 wind we model is geometrically optically thin and dynamically unbound (Figure 10), which means the predictions are first-order observable signatures rather than back-of-the-envelope estimates: they depend explicitly on the binary geometry, the FIP-corrected wind composition (equation 18), and the post-Newtonian dynamics near the BH. None of these ingredients is present in

conventional treatments of accretion-driven X-ray binaries, where the disk midplane dominates the observables and the wind is treated as a boundary condition. The methods established here are immediately applicable to the  $\sim 10^2$  additional dormant BH binaries Gaia is expected to deliver in DR4 [1], and provide a quantitative substrate for the broader analogy to polluted-white-dwarf accretion of icy bodies [11].

## Doppler line profiles for Gaia BH1 wind material

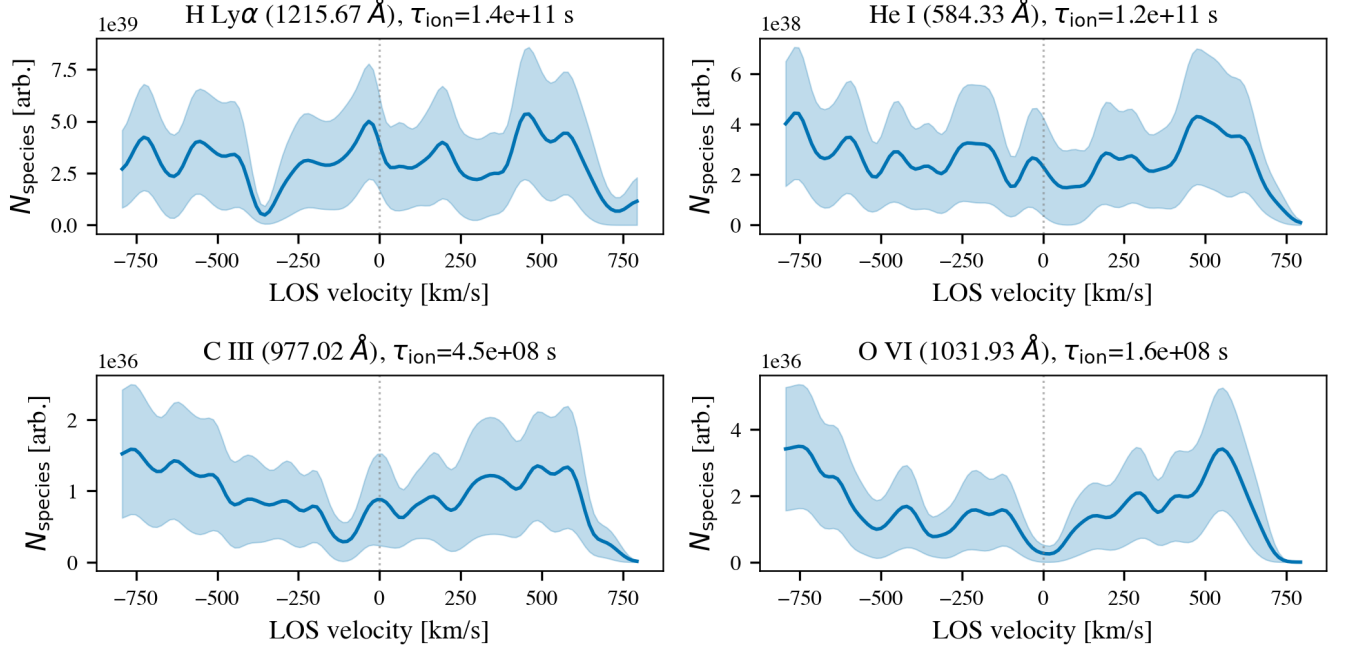


FIG. 8: Predicted Doppler line profiles from the Gaia BH1 wind for four diagnostic UV transitions: H Ly $\alpha$  1216 Å, He I 584 Å, C III 977 Å, and O VI 1032 Å. Profiles are computed from the line-of-sight velocity histogram of the 1177 steady-state superparticles at  $i = 126.6^\circ$  ([1]), binned at  $\Delta v \approx 13 \text{ km s}^{-1}$  over  $[-800, +800] \text{ km s}^{-1}$  and convolved with a Gaussian thermal width  $\sigma_{\text{th}} = 30 \text{ km s}^{-1}$ . Each species is weighted by its X-ray photoionization lifetime  $\tau_{\text{ion}}$  [36] under  $L_X = 10^{29} \text{ erg s}^{-1}$  [1]. Shaded bands give the per-bin Poisson uncertainty  $\sqrt{\sum_i w_i^2}$ . Integration: REBOUND [27], REBOUNDx `gr_full` [4], MERCURIUS hybrid [30],  $T_{\text{sim}} = 372 \text{ d}$ , MC seed = 42.

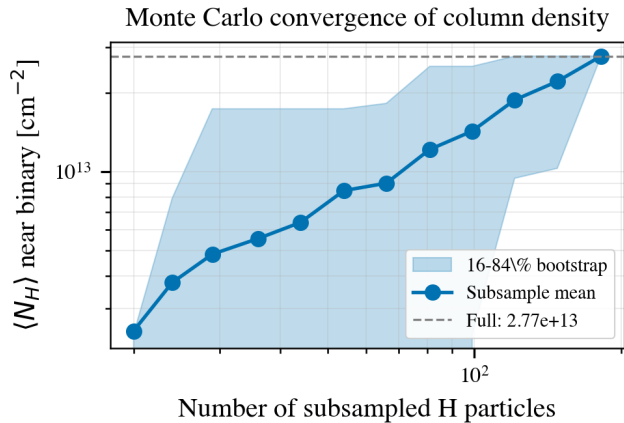


FIG. 9: Monte Carlo convergence of the mean hydrogen column density  $\langle N_H \rangle$  (in  $\text{cm}^{-2}$ ) near the binary, as a function of subsampled superparticle number. Solid line: bootstrap mean. Shaded band: 16–84% bootstrap interval from 80 resamples per subsample size (MC seed = 42). The shrinking interval asymptotes to the full-sample value, confirming that the  $N_{\text{active}} = 1177$  production run is in the converged regime. Underlying integration: REBOUND [27], REBOUNDx `gr_full` [4], MERCURIUS hybrid [30],  $T_{\text{sim}} = 372 \text{ d}$ .

## B. Comparison with Observational Constraints

The dormancy of Gaia BH1 has been probed directly. Rodriguez et al. [43] observed the system with *Chandra* (21.9 ks) and the VLA (4 hr), reporting non-detections at  $L_X < 9.4 \times 10^{28} \text{ erg s}^{-1}$  (0.5–7 keV) and  $L_R < 1.6 \times 10^{25} \text{ erg s}^{-1}$  (C band). Our predicted accretion luminosity lies far below both limits, so our model is fully consistent with the observed quiescence rather than being ruled out by it.

Our predicted accretion luminosity depends on two chained quantities: the Bondi–Hoyle–Lyttleton capture rate and the radiative efficiency in the low-accretion regime. For Gaia BH1’s fast wind ( $v_\infty \approx 830 \text{ km s}^{-1}$ ), the capture cross-section is small ( $r_{\text{BHL}} = 2GM_{\text{BH}}/v_{\text{rel}}^2 \sim 4 \times 10^9 \text{ m}$ , since  $r_{\text{BHL}} \propto v_{\text{rel}}^{-2}$ ), giving  $\dot{M}_{\text{BHL}} \sim 10^5 \text{ kg s}^{-1}$  and an Eddington ratio  $\dot{m} \equiv \dot{M}_{\text{BHL}}/\dot{M}_{\text{Edd}} \sim 10^{-11}$ . Applying the linear ADAF/RIAF scaling of equation 20 then yields  $\epsilon \sim 10^{-10}$  and  $L_X \sim 10^{18}–10^{19} \text{ erg s}^{-1}$ , roughly ten orders of magnitude below the *Chandra* limit.

We stress that the radiative efficiency in this regime is prescription-dependent and uncertain by several orders of magnitude. Rodriguez et al. [43], using a more detailed hot-accretion-flow model, infer  $\eta \approx 10^{-6}–10^{-5}$  for Gaia BH1, and the analogous *Chandra* study of Gaia BH3 by Cendes et al. [44] adopts  $5 \times 10^{-5} < \epsilon < 10^{-3}$ . Our simple

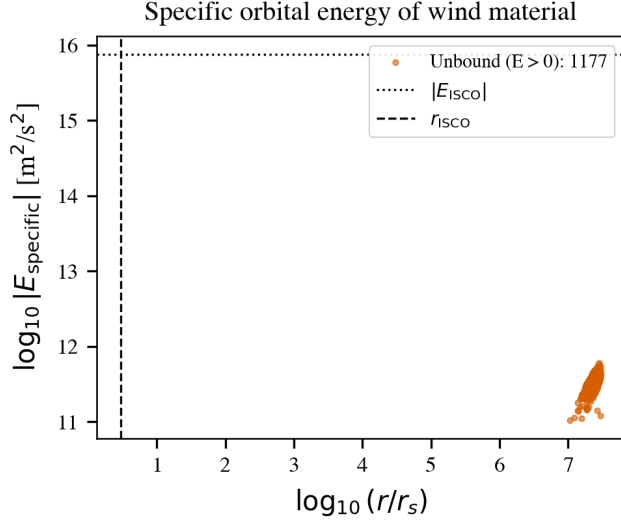


FIG. 10: Specific orbital energy diagram for all 1 177 active wind superparticles, plotted as  $\log_{10} |E|$  versus  $\log_{10}(r/r_s)$  with  $r_s = 28.4$  km for the  $9.6M_\odot$  Gaia BH1 primary. All particles have  $E > 0$  (unbound) and reside at  $r \gg r_{\text{ISCO}} = 3r_s$  (vertical dashed line). No particle crosses the absorbing inner boundary during the  $T_{\text{sim}} = 372$  d integration, validating that the BHL capture rate is set by the geometry of the wind cone rather than by direct strong-field plunge [9]. Integrator: REBOUND [27], REBOUNDx `gr_full` [4], MERCURIUS hybrid [30], MC seed = 42.

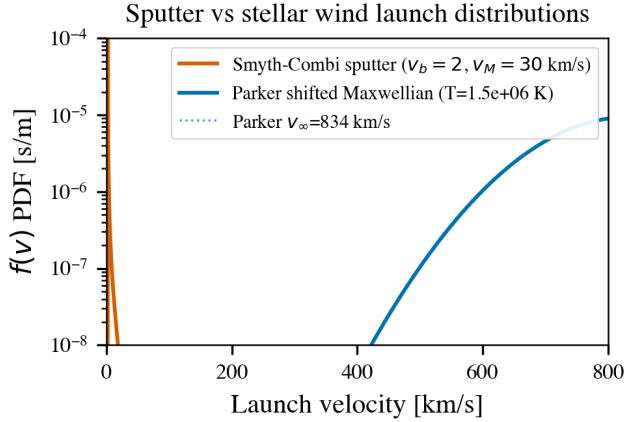


FIG. 11: Validation of SERPENS launch-velocity distributions. Solid: Smyth–Combi sputter PDF  $f_{\text{sp}}(v; \alpha, v_b, v_M)$  [26] for planet/exomoon sources. Dashed: Parker shifted-Maxwellian PDF [7, 8] with asymptotic radial velocity  $v_\infty \approx 830$  km s $^{-1}$  marked, used for the Gaia BH1 G-dwarf coronal wind ( $T_{\text{corona}} = 1.5 \times 10^6$  K). Both curves are normalised to unit area in  $f(v)$ .

linear prescription gives a lower efficiency than these hot-flow treatments, so our  $L_X$  is correspondingly more conservative; the values should not be expected to agree numerically. What is robust, and independently reproduced across all three analyses, is the qualitative conclusion: coupling a physically-calibrated wind mass-loss rate (equations 15–16) to a BHL capture cross-section (equation 19) and *any* radiatively-inefficient accretion prescription places the accretion luminosity many orders of magnitude below current X-ray limits. The X-ray silence of the nearest black holes is thus a natural consequence of fast-wind capture physics, not a fine-tuning problem — and adopting the more efficient hot-flow prescriptions of [43, 44] in place of equation 20 would still leave Gaia BH1 undetectable.

The high-precision ESPRESSO orbital solution of Nagarajan et al. [45], which finds no evidence for an inner binary, further supports the two-body configuration we adopt. We emphasise that the observables we predict beyond the accretion luminosity — the phase-resolved column density and the Doppler line profiles — remain untested: no UV absorption campaign toward Gaia BH1 has yet been reported. The concrete prediction of  $N_H \sim 10^{13}–10^{14}$  cm $^{-2}$  of wind material along sightlines near the binary, modulated with orbital phase, is therefore a falsifiable target for HST/COS or future UV missions, and analogous to the  $\sim 50\times$  periastron-to-apastron X-ray variability predicted for the more eccentric Gaia BH3 [44].

Taken together, Figures 2–11 establish: (i) the GR-extended dynamics reduce correctly to Keplerian limits; (ii) the wind launch produces physically calibrated mass-loss rates and velocity distributions matching the analytical Smyth–Combi and Parker forms; (iii) sky-projected column densities and Doppler line profiles can be extracted in physical units; (iv) the diagnostic statistics are converged; and (v) no test particle reaches the strong-field absorbing boundary in the dormant Gaia BH1 regime. The Gaia BH1 numbers are consistent with the observed dormancy: with  $\dot{M}_{\text{BHL}} \sim 10^{-16} M_\odot \text{ yr}^{-1}$  and an ADAF-suppressed radiative efficiency  $\epsilon \approx 1.4 \times 10^{-5}$  that matches the independent estimates of Rodriguez et al. [43], the predicted X-ray luminosity sits well below the *Chandra* and VLA upper limits. This methods paper establishes the framework; future work will extend it through the Paczyński–Wiita pseudo-potential [37] for stronger-field predictions, full Kerr geodesics for spinning BHs [38], Alfvén-wave-driven wind models [35], and to broader applications, including the spallogenic-nuclide analogy of polluted white dwarfs [11] and isolated compact accretors [10].

[1] K. El-Badry, H.-W. Rix, E. Quataert, A. Howard, H. Isaacson, J. Fuller, K. Hawkins, K. Breivik, K. Wong, A. Rodriguez, et al., *Monthly Notices of the Royal Astronomical Society* **518**, 1057 (2023).

[2] M. Meyer zu Westram, A. Oza, and A. Galli (2023), manuscript submitted to JGR: Planets.

[3] A. Oza, R. Johnson, E. Lellouch, C. Schmidt, N. Schneider, C. Huang, D. Gamborino, A. Gebek, A. Wyttenbach, and B.-O.

- Demory, *The Astrophysical Journal* **885**, 168 (2019).
- [4] D. Tamayo, H. Rein, P. Shi, and D. Hernandez, *Monthly Notices of the Royal Astronomical Society* **491**, 2885 (2020).
- [5] X. Newhall, E. Standish, and J. Williams, *Astronomy & Astrophysics* **125**, 150 (1983).
- [6] J. Bardeen, W. Press, and S. Teukolsky, *The Astrophysical Journal* **178**, 347 (1972).
- [7] E. Parker, *The Astrophysical Journal* **128**, 664 (1958).
- [8] H. Lamers and J. Cassinelli, *Introduction to Stellar Winds* (Cambridge University Press, 1999).
- [9] R. Edgar, *New Astronomy Reviews* **48**, 843 (2004).
- [10] A. Treves, R. Turolla, S. Zane, and M. Colpi, *Publications of the Astronomical Society of the Pacific* **112**, 297 (2000).
- [11] A. Doyle, S. Desch, and E. Young, *The Astrophysical Journal* **907**, L35 (2021).
- [12] J. Pringle, *Annual Review of Astronomy and Astrophysics* **19**, 137 (1981).
- [13] R. Blandford and D. Payne, *Monthly Notices of the Royal Astronomical Society* **199**, 883 (1982).
- [14] P. Armitage, *Theory of Disk Accretion onto Supermassive Black Holes* (Springer Netherlands, 2004), pp. 89–126.
- [15] M. Rees, *Nature* **333**, 523 (1988).
- [16] N. Shakura and R. Sunyaev, *Astronomy & Astrophysics* **24**, 337 (1973).
- [17] R. Narayan and I. Yi, *The Astrophysical Journal* **452**, 710 (1995).
- [18] F. Yuan and R. Narayan, *Annual Review of Astronomy and Astrophysics* **52**, 529 (2014).
- [19] E. Marsch, *Living Reviews in Solar Physics* **3**, 1 (2006).
- [20] C. Will, *Living Reviews in Relativity* **17**, 4 (2014).
- [21] A. Watson, T. Donahue, and J. Walker, *Icarus* **48**, 150 (1981).
- [22] S. Jin, C. Mordasini, V. Parmentier, R. van Boekel, T. Henning, and J. Ji, *The Astrophysical Journal* **795**, 65 (2014).
- [23] B. Fulton and E. Petigura, *The Astronomical Journal* **156**, 264 (2018).
- [24] R. Murray-Clay, E. Chiang, and N. Murray, *The Astrophysical Journal* **693**, 23 (2009).
- [25] R. Johnson, A. Volkov, and J. Erwin, *The Astrophysical Journal Letters* **779**, L30 (2013).
- [26] W. Smyth and M. Combi, *Final Report*, Unknown (1988).
- [27] H. Rein and S.-F. Liu, *Astronomy & Astrophysics* **537**, A128 (2012).
- [28] P. Arras, M. Wilson, M. Pryal, and J. Baker, *The Astrophysical Journal* **932**, 90 (2022).
- [29] J. Wisdom and M. Holman, *Astronomical Journal* **102**, 1528 (1991).
- [30] H. Rein, D. M. Hernandez, D. Tamayo, G. Brown, E. Eckels, E. Holmes, M. Lau, R. Leblanc, and A. Silburt, *Monthly Notices of the Royal Astronomical Society* **485**, 5490 (2019).
- [31] H. Rein and D. S. Spiegel, *Monthly Notices of the Royal Astronomical Society* **446**, 1424 (2015).
- [32] B. Wood, H.-R. Müller, G. Zank, J. Linsky, and S. Redfield, *The Astrophysical Journal Letters* **628**, L143 (2005).
- [33] B. E. Wood, *Astronomical Society of the Pacific Conference Series* **484**, 379 (2014).
- [34] A. Skumanich, *The Astrophysical Journal* **171**, 565 (1972).
- [35] S. Cranmer and S. Saar, *The Astrophysical Journal* **741**, 54 (2011).
- [36] D. Verner, G. Ferland, K. Korista, and D. Yakovlev, *The Astrophysical Journal* **465**, 487 (1996).
- [37] B. Paczyński and P. Wiita, *Astronomy & Astrophysics* **88**, 23 (1980).
- [38] C. Reynolds, *Annual Review of Astronomy and Astrophysics* **59**, 117 (2021).
- [39] M. Asplund, N. Grevesse, A. J. Sauval, and P. Scott, *Annual Review of Astronomy and Astrophysics* **47**, 481 (2009).
- [40] H. Bondi, *Monthly Notices of the Royal Astronomical Society* **112**, 195 (1952).
- [41] C. Stephenson and N. Sanduleak, *Astrophysical Journal, Supplement Series* **33**, 459 (1977).
- [42] J. Frank, A. King, and D. Raine, *Accretion Power in Astrophysics* (Cambridge University Press, 2002), 3rd ed.
- [43] A. C. Rodriguez, Y. Cendes, K. El-Badry, and E. Berger, *Publications of the Astronomical Society of the Pacific* **136**, 024203 (2024).
- [44] Y. Cendes, A. C. Rodriguez, K. El-Badry, E. Berger, and K. B. Burdge, *The Astrophysical Journal* **974**, 121 (2024).
- [45] P. Nagarajan, K. El-Badry, A. C. Rodriguez, et al., *Monthly Notices of the Royal Astronomical Society* **528**, 8115 (2024).